

**PART 4:** Perform a Simple Length-Filtering Experiment to test how much of the genome assembly is made up of short sequences versus long ones by removing all sequences under a certain length and seeing how the statistics change.

**SPECIES:** *Cynopterus brachyotis* (lesser short-nosed fruit bat) genome assembly (GCA\_009793145.1)

**GALAXY FILE NAME:** 5: *Cynopterus\_brachyotis\_filtered\_10kb*

**NOTE:** This is only a screenshot as I cannot upload the raw file on GitHub due to its large file size. Therefore, this screenshot will serve as evidence that I indeed run the data through *Filter sequences by length*.

The screenshot displays the Galaxy web interface. On the left, the 'Tools' sidebar is visible, categorized into 'General Text Tools' (including Text Manipulation, Filter and Sort, Join, Subtract and Group, and Datamash) and 'Genomic File Manipulation' (including FASTA/FASTQ, FASTQ Quality Control, SAM/BAM, BED, VCF/BCF, and Nanopore). The main panel shows a dataset titled '5: Cynopterus\_brachyotis\_filtered\_10kb' with 12,483 sequences in FASTA format, totaling 1.7 GB. A warning message states: 'This dataset is large and only the first 100 KB is shown below.' Below this, a preview of the FASTA sequence is displayed, starting with:   
>SSHV01000001.1 Cynopterus brachyotis isolate CB-01  
scaffold237\_cov87, whole genome shotgun sequence  
AATTACTTTTCTCCCTACCACTGCAACCACCTTCTTAGAGAAACTC  
GTCAACTGATTGGAGACCAGGGAAGGACACCTGATTCACATTTC  
GGGAATATTGCCTGAGACAAATCGCTACAGGGATCAAACCTGGAC  
CAGTTGTGGGAATCTAGACTACAGAGAGAAGAATAATGTAGACC  
GGACAAAATTGGCTGCATTAGGGCTTTCCAGAAGCCAGGCATTI  
attcttcttcttcttcttcttcttcttcttcttcttcttcttcttcttctt  
cctacttcttcttcttcttcttcttcttcttcttcttcttcttcttcttctt  
TCTGTACCAAAAAGcaaaaaacttaaaaaattatgcaaaaaatatatgctaactTTACTCA  
taaaaaattactttttaagcaatttgaAGTTACTCTTATTTCCAAAGTATTTCTAA  
GGTAATTGATTACATGGGTTACACTTAGTCTGAGTAACATGTT.  
gcatafaaaaagttaatttaagcaaatgtcttgcatttaaatattagaTCTAAAtatggatntaafattttg  
ggACAGAACATCtggaaataatagattttaaggCCTATGATAGTCTTAAAACTI  
CTGGAGTGGTGTACATAAAGCATGCTGGACCTGAAATGACATC

On the right, the 'History' panel shows a list of datasets, with '5: Cynopterus\_brachyotis\_filtered\_10kb' selected. Other datasets in the history include 'Genome\_Exploration\_II\_OBINET A\_Cynopterus\_brachyotis', '6: Cynopterus\_brachyotis\_Filtered\_10kb\_Assembly\_Statistics\_gfastats', '4: Cynopterus\_brachyotis\_Histogram', '3: Cynopterus\_brachyotis\_Compute sequence length', '2: Cynopterus\_brachyotis\_Assembly\_Statistics\_gfastats', and '1: Cynopterus\_brachyotis\_genome\_origin\_alma.gz'.